

Linear Regression

Equation of a line, 3D & Hyperplane:

⇒ What is a line? A line is a straight path that extends indefinitely in both directions.

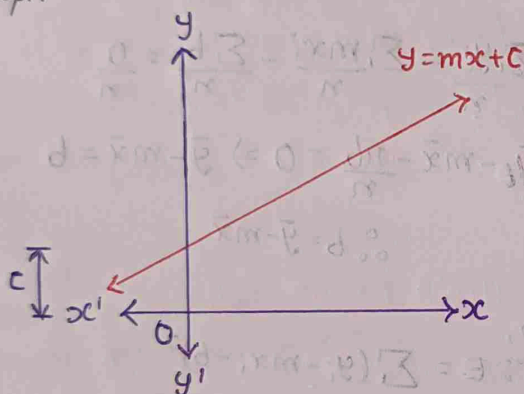
⇒ General eqⁿ of a line:

$$y = mx + c$$

where, $m = \text{slope} = \frac{\Delta x}{\Delta y}$

$c = y\text{-intercept}$

⇒ On graph:



⇒

No. of Features	Geometry	Equation
1	Line	$\hat{y} = w_1 x_1 + b$
2	Plane	$\hat{y} = w_1 x_1 + w_2 x_2 + b$
3	Hyperplane (in 4D)	$\hat{y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$
n	Hyperplane	$\hat{y} = w_1 x_1 + \dots + w_n x_n + b$

Distance of a point from a plane:

⇒ Equation of a plane:

$$Ax + By + Cz + D = 0$$

where,

- A, B, C determine the orientation of the plane.
- D determines its position.

⇒ Suppose the point is, $P(x_0, y_0, z_0)$, then the distance from P to the plane is:

$$d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

Simple Linear Regression:

⇒ What is linear regression? It's a supervised learning algorithm used to predict a continuous numerical output.

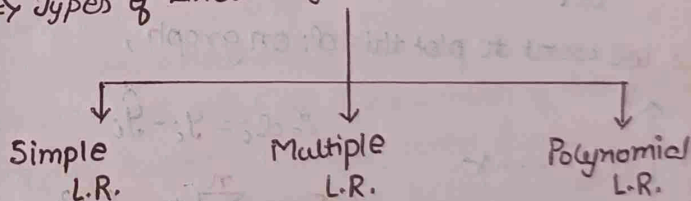
⇒ Why supervised? Bcz during training, the model sees:

Input $x \rightarrow$ Known Output y
and learns the relationship b/w them.

⇒ Why is it called Simple Linear Regression? One input feature $\rightarrow$ one continuous output.

⇒ What is the Best Fit Line? The line which represents the data as accurately as possible. The model tries to find the line where the errors are as small as possible.

⇒ Types of Linear Regression:



⇒ In ML, we usually write:

$$\hat{y} = mx + b$$

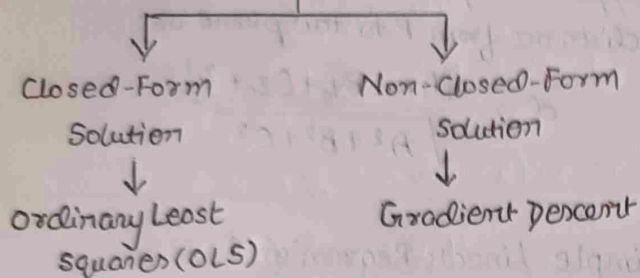
Here, m is called weight. The weight tells us how much the predicted output changes for a one-unit inc. in the input.

And, b is the offset.

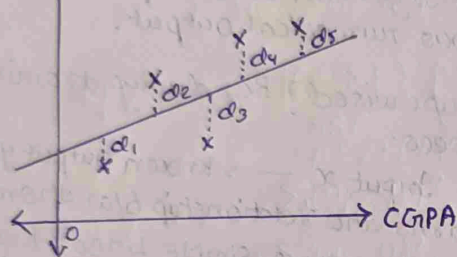
⇒ Prediction = Effect of Input + Offset

⇒ How to find the value of 'm' and 'b'?

There are two ways to do so:



⇒ Package



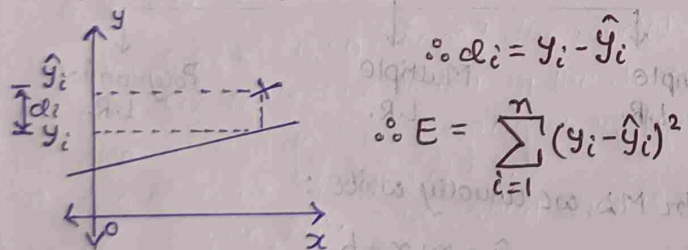
Clearly, total Error = $d_1 + d_2 + \dots + d_n$

∵ some error can be -ve, so we take square.

$$E = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

$$= \sum_{i=1}^n d_i^2 \rightarrow \text{Error Function}$$

Now, we want to plot this d_i on graph,



$$\hat{y}_i = mx_i + b$$

$$E(\text{of terms of } m \text{ and } b) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

∵ our aim is to minimise the error, and 'm' and 'b' the variable.

We know,

$$y = f(x)$$

Let, $b = 0$ (constant)

$$E(m) = \sum_{i=1}^n (y_i - mx_i)^2$$

Similarly, $m = \text{constant}$ (let)

$$E(b) = \sum_{i=1}^n (y_i - x_i - b)^2$$

Now,

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b} \sum_{i=1}^n (y_i - mx_i - b)^2 = 0$$

$$\Rightarrow \sum \frac{\partial}{\partial b} (y_i - mx_i - b)^2 = 0$$

$$\Rightarrow \sum (-2)(y_i - mx_i - b) = 0 \Rightarrow \sum (y_i - mx_i - b) = 0$$

$$\Rightarrow \sum y_i - \sum mx_i - \sum b = 0$$

$$\Rightarrow \frac{\sum y_i}{n} - \frac{\sum mx_i}{n} - \frac{\sum b}{n} = 0$$

$$\Rightarrow \bar{y} - m\bar{x} - \frac{n b}{n} = 0 \Rightarrow \bar{y} - m\bar{x} = b$$

$$\therefore b = \bar{y} - m\bar{x}$$

Now,

$$\therefore E = \sum (y_i - mx_i - b)^2$$

$$\Rightarrow E = \sum (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$\frac{\partial E}{\partial m} = \sum (2)(y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$\Rightarrow \sum (y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$\Rightarrow m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\therefore m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Regression Metrics:

⇒ Most common ones are:

① MAE

② MSE

③ RMSE

④ R^2

⑤ Adjusted R^2

How wrong are my predictions?

How well does the model explain the target?

⇒ Mean Absolute Error (MAE): It tells us the average diff. b/w actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

e.g.,

Actual: [10, 20, 30] | where, y_i = actual value

Errors: [-2, 2, 3] | $\hat{y}_i$ = predicted value

Predicted: [12, 18, 33] | n = no. of observations

Absolute errors: [2, 2, 3]

$$\therefore MAE = \frac{2+2+3}{3} = 2.33$$

So, we can say, on average, the model's prediction are off by about 2.33 units.

∴ Smaller MAE → Better

Also, advantage:

- Same unit as target.
- Robust to outliers, as effect is less as compared to MSE/RMSE.
- Every error contributes proportionally.

Also, disadvantage:

- Doesn't strongly penalize large errors.

⇒ Mean Squared Error (MSE): It tells us the average squared prediction error.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

e.g., Suppose errors are, 2, -2, 3

$$\text{Then, } MSE = \frac{(2)^2 + (-2)^2 + (3)^2}{3} = 5.67$$

∴ Smaller MSE → Better

Also, advantage:

- Strongly penalizes large errors.
- Differentiable

Also, disadvantage:

- Difficult to interpret

⇒ Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

∴ RMSE small → Better

Also, advantages:

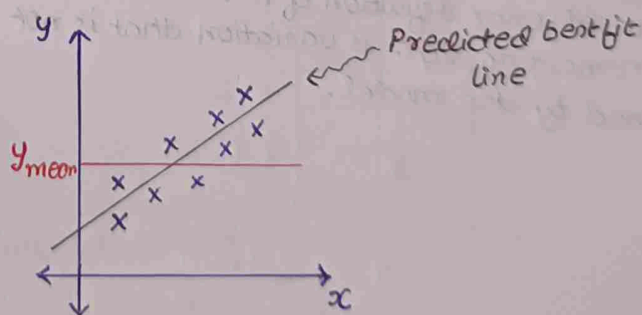
- Same unit as target.
- Penalize large errors.

Also, disadvantages:

- Sensitive to outliers.

⇒ R^2 score: R^2 asks, how much of the variation in the target does my model explain?

Also, called coefficient of Determination.



∴ R^2 uses two quantities :

① Total sum of squares (SST) (also called TSS)

It measures the total variation of the actual values around their mean :

$$SST = \sum (y_i - \bar{y})^2$$

It answers, how much variation exists in the target is total ?

② Residual sum of squares (SSR) (or SSE)

$$SSR = \sum (y_i - \hat{y}_i)^2$$

It measures, how much variation / error is still left unexplained by the regression model.

$$R^2 = 1 - \frac{SSE}{SST}$$

$$= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$= 1 - \frac{\text{unexplained variation}}{\text{total variation}}$$

Also, R^2 closer to 1 is better.

Also, $R^2 = 0 \Rightarrow SSE = SST$

$R^2 = 1 \Rightarrow SSE = 0$

Also, $R^2 < 0 \Rightarrow$ worse than the mean baseline

∴ what does $R^2 = 0.80$ means? The model explains 80% of the variation in the target variable relative to the mean baseline, based on squared-error definition of R^2 .

The remaining 20% is variation that is not explained by the model.

⇒ Adjusted R^2 :

Since, R^2 already existed then why we need adjusted R^2 ? ordinary R^2 artificially inc. or stays the same whenever you add new feature to a model. This flaw allows irrelevant variables to make the model look better without providing genuine predictive value.

Adjusted R^2 solves this by introducing a penalty for adding predictors. A common formula is :

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \frac{n - 1}{n - p - 1}$$

where,

n = no. of observations

p = no. of predictor / input features

But how does it helps? It asks, "did anything this feature improve the model enough to justify the additional complexity?"

Yes: Adjusted $R^2 \uparrow$

No: Adjusted $R^2 \downarrow$

Multiple Linear Regression (MLR)

⇒ What is MLR? It is a L.R. where we use more than one input feature / column to predict one continuous target / output.

⇒ 2 inputs → Plane

⇒ The model is written as :

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Predicted value

⇒ What do we need to find? The coefficients / parameters: $\beta_0, \beta_1, \beta_2$

These are what the model learns from the training data.

Specifically,

- $\beta_0 \rightarrow$ intercept / bias
- $\beta_1 \rightarrow$ coefficient / weight of x_1
- $\beta_2 \rightarrow$ coefficient / weight of x_2

$\Rightarrow$ Equation for n input features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_n x_n$$

$$= \beta_0 + \sum_{j=1}^n \beta_j x_j$$

where, $n =$ no. of input features
 $x_j = j$ th feature
 $\beta_j =$ weight corresponding to that feature
 $\beta_0 =$ bias

$\Rightarrow$ What do these coefficients mean?

Suppose,

$$LPA = \beta_0 + \beta_1 (CGPA) + \beta_2 (IQ)$$

e.g.,

$$LPA = -5 + 3(CGPA) + 0.05(IQ)$$

$$\therefore \beta_0 = -5, \beta_1 = 3, \beta_2 = 0.05$$

Clearly, $\beta_1 = 3$ i.e., coeff. of CGPA is 3. It means, for a 1 unit inc. in CGPA (say 8 $\rightarrow$ 9), the predicted LPA inc. by 3 units, keeping IQ constant (say IQ = 120).

Also, $\beta_2 = 0.05$, it means for every 1 unit inc. in IQ, predicted LPA inc. by 0.05 holding CGPA constant. So, if IQ inc. from 120 $\rightarrow$ 130, the predicted LPA inc. by: $0.05(10) = 0.5$.

$\Rightarrow$ Also, keep in mind comparing $\beta_1 = 3$ and $\beta_2 = 0.05$ is not a valid idea as magnitude of the coeff. depends on the scale/units of the features.

$\Rightarrow$ What about β_0 ? It's the intercept req. to position the regression plane appropriately.

$\Rightarrow$

	CGPA	IQ	Gender	LPA
Predicted LPA	x_1	x_2	x_3	$y \leftarrow$ actual LPA

$$\therefore \hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

Now, let there be 100 students having below columns as above.

$$\therefore \hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_{100} \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \beta_3 x_{13} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \beta_3 x_{23} \\ \vdots & \vdots & \vdots & \vdots \\ \beta_0 & \beta_1 x_{1001} & \beta_2 x_{1002} & \beta_3 x_{1003} \end{bmatrix}$$

(capital Y)

Let we have m columns, n rows,

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \dots & \beta_m x_{1m} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \dots & \beta_m x_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \beta_0 & \beta_1 x_{n1} & \beta_2 x_{n2} & \dots & \beta_m x_{nm} \end{bmatrix}$$

$$\Rightarrow \hat{y} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1m} \\ 1 & x_{21} & \dots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nm} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$$

$$\Rightarrow \hat{y} = X\beta$$

where, $X =$

	x_1	x_2	x_3
1	x_{11}	x_{12}	x_{13}
1	x_{21}	x_{22}	x_{23}
1	x_{31}	x_{32}	x_{33}
$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	x_{n1}	x_{n2}	x_{n3}

(for 3 col and n rows)

Also, let matrix of output column be y ,

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Now, let $E = Y - \hat{Y}$

$$= \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

From simple LR, we know,

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

By properties,

$$\begin{aligned} E &= E^T E \quad (\text{calculated}) \\ &= (Y - \hat{Y})^T (Y - \hat{Y}) \\ &= (Y^T - \hat{Y}^T)(Y - \hat{Y}) \\ &= [Y^T - (X\beta)^T](Y - X\beta) \quad [\because \hat{Y} = X\beta] \\ &= Y^T Y - Y^T X\beta - (X\beta)^T Y + (X\beta)^T X\beta \\ &= Y^T Y - 2Y^T X\beta + \beta^T X^T X\beta \\ &\quad [\because Y^T X\beta = (X\beta)^T Y] \end{aligned}$$

$$\therefore \frac{dE}{d\beta} = \frac{d}{d\beta} [Y^T Y - 2Y^T X\beta + \beta^T X^T X\beta]$$

$$\Rightarrow = 0 - 2Y^T X + \frac{d}{d\beta} (\beta^T X^T X\beta)$$

$$\Rightarrow = -2Y^T X + 2X^T X\beta^T = 0$$

$$\Rightarrow 2X^T X\beta^T = 2Y^T X$$

$$\Rightarrow \beta^T = Y^T X (X^T X)^{-1}$$

$$\Rightarrow (\beta^T)^T = [Y^T X (X^T X)^{-1}]^T = [(X^T X)^{-1}]^T (Y^T X)^T$$

$$\Rightarrow \beta = (X^T X)^{-1} X^T Y$$

where,

$$X \rightarrow X_{\text{train}}$$

$$Y \rightarrow Y_{\text{train}}$$

Main Assumptions of LR :

$\Rightarrow$ These assumptions tell us when LR is appropriate. Since not every assumption is required merely to fit a regression model.

Assumption	Simple Meaning
Linear Relationship	Inputs & target have an appropriate linear relationship.
No multicollinearity	Predictors aren't excessively correlated with each other.
Normal Residuals	Residuals approx follow a normal distribution.
Homoscedasticity	Residual variance is roughly constant
No autocorrelation	Errors are not systematically related to each other.